

A Validated Camera-to-Actuator Pipeline for Vision-Based Traffic Congestion Analytics and Beacon-Triggered Emergency Priority

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Abstract—Fixed-time and loop-actuated traffic signals cannot measure the traffic they control: a loop reports presence, not condition, and a vehicle count cannot separate an empty road from a standstill. This paper reports a complete measurement chain from a single roadside monocular camera to physical roadside actuators, in which every stage is validated rather than asserted. Vehicles are detected and tracked, and their image motion is converted to true ground speed through a road-plane homography anchored at the tyre contact patch and timed against a media clock that is independent of processing rate. Speeds and passenger-car-equivalent density are condensed into a bounded congestion index with presence and emptiness guards, and mapped to a three-level state through a hysteretic comparator. Emergency vehicles are recognised by two complementary cues, with intervention authorised only by a confirmed flashing beacon detected as a temporal switching signal rather than by vehicle livery. Against constructed ground truth the speed pipeline returns 18.00, 36.00 and 72.00 km/h for targets scripted at 18, 36 and 72 km/h, and an identical 16.2 km/h across four playback rates. Over 26 071 logged seconds and 56 098 vehicle records, mean measured speed falls monotonically from 68.3 to 5.4 km/h across the index range. Making the beacon the trigger reduced night-time false alarms from 29% of all traffic to zero. The pipeline sustains 22–28 frames/s on a laptop-class GPU. Every claim is bounded by the evidence class that supports it.

Index Terms—Adaptive signal control, computer vision, congestion measurement, edge computing, emergency vehicle priority, homography, intelligent transportation systems, object detection, speed estimation, vehicle tracking.

I. INTRODUCTION

URBAN traffic control is a sensing problem before it is an actuation problem. A fixed-time plan can allocate green time, but it cannot establish whether an approach is empty, whether a queue is forming, or whether an emergency vehicle is requesting priority: its timings derive from a survey conducted at some point in the past, so it spends green time on roads with nothing to discharge while traffic accumulates elsewhere. Vehicle-actuated control improves on this using inductive loops, but a loop reports that metal is above it, not how many vehicles there are, how fast they are moving, or what class they belong to; loops also require the carriageway to be cut, and they fail invisibly.

A camera is already common roadside infrastructure, needs no intervention in the carriageway, and carries far more information. What historically prevented that information from being used for control was that extracting it in real time was computationally out of reach—a constraint that has now lifted, since one-stage detectors run at video rate on hardware small enough for a roadside cabinet [1], [2].

A. Problem Statement

The problem addressed here is the gap between what the road is doing and what the signal knows about it. Four consequences follow, and each was treated as a requirement.

- 1) *Congestion cannot be quantified from a count alone.* A low count occurs both on an empty road and in a standstill, because a jammed road discharges few vehicles past any point; a system reporting counts has measured flow, which is ambiguous, not congestion.
- 2) *Speed cannot be recovered without geometry.* Under perspective projection the same physical velocity produces very different pixel velocities depending on range, so reporting pixel displacement per second as a speed yields numbers that cannot be acted upon.
- 3) *Emergency vehicles are treated as ordinary traffic,* and an ambulance on a call is delayed by exactly the congestion the system is meant to manage.
- 4) *The operator is left outside the loop.* An operator who cannot see what the system measures, cannot distinguish live from stale data and cannot override a decision, cannot be held responsible for the junction.

B. Contributions

The contribution is integrative and evidential rather than algorithmic: the individual techniques are mature, but a prototype in which the whole chain is closed and every link validated is rare.

- A metric speed pipeline converting image motion to kilometres per hour through a road-plane homography anchored at the tyre contact patch, decoupled from processing rate by a media clock, and validated against constructed ground truth to two decimal places (Sections IV, IX).
- A bounded congestion index combining speed and passenger-car-equivalent density, guarded so that neither a single slow vehicle nor an empty road registers as

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congestion, with a hysteretic level mapping suitable for driving physical actuators (Section V).

- An emergency decision separating what a vehicle *is* from what it *is doing*, authorising intervention only on a confirmed two-state beacon—removing a systematic fault affecting 29% of night traffic (Section VI).
- A detector–resolution operating point selected by measurement on the deployment footage rather than by published benchmark, overturning two initial assumptions (Section IV-A).
- An end-to-end demonstrator driving three physical actuators, evaluated over 26 071 logged seconds (Section IX), with an explicit evidence-boundary framework (Table IV) stating which claims each class of test supports and which it does not.

Section II reviews related work and Section III the architecture; Sections IV–VIII present the pipeline, index, emergency logic, interface and actuators; Section IX reports results, Section X the limitations, and Section XI concludes.

II. RELATED WORK

A. Traffic Signal Control

Signal control divides into three generations. Fixed-time control applies a pre-computed plan; it is cheap, predictable and blind. Vehicle-actuated control extends or truncates a stage in response to a detector, at the cost of civil work and a sensor reporting only presence. Adaptive systems such as SCOOT [3] and SCATS [4] optimise cycle time, split and offset across a network and remain the state of practice. The literature is consistent that the limiting factor is the coverage and quality of the detection layer rather than the optimisation formulation. That observation is the premise of this work: the detection layer is replaced, not the optimiser.

B. Quantifying Congestion

Traffic engineering does not define congestion by vehicle count. The *Highway Capacity Manual* [5] grades a facility by level of service, derived from density and operating speed rather than flow, precisely because a low count occurs both on an empty road and in a standstill; the speed–density relationship of Greenshields [6] captures the same insight. Two conventions are adopted directly: passenger car equivalents (PCE), which weight vehicles by the road space they consume, and temporal stability, since a metric driving a physical actuator oscillates whenever a raw threshold comparison sits on a boundary—addressed in control engineering by hysteresis.

C. Vision-Based Traffic Monitoring

Vision-based monitoring has moved through background subtraction, feature-and-classifier methods, and convolutional detectors, which replaced both. These divide into two-stage detectors, of which Faster R-CNN [7] is canonical, and one-stage detectors such as SSD [2] and the YOLO family [1], [8]. The trade-off is not close here: measuring speed requires the same vehicle to be localised repeatedly within a short window, which makes frame rate a functional requirement

rather than a convenience. Surveys of computer vision in intelligent transportation [9], [10] confirm both the maturity of the detection layer and the scarcity of work closing the loop onto control.

Table I compares seven representative works. Two patterns are visible. Every work reporting a high detection accuracy stops at detection; of the seven, only three close the loop onto signal control, and each of those was validated on a single intersection or in simulation. None reports a measured speed in engineering units, and none publishes its measurement to an operator interface in real time.

D. Emergency Priority and Operator Interfaces

Pre-emption systems in service are largely transmitter-based: the emergency vehicle carries an emitter and the junction responds. These are reliable but require equipment in every vehicle and at every junction. Sensor-based alternatives divide into audio detection of the siren sweep, robust to occlusion and darkness but degraded by traffic noise, and vision detection, which either classifies emergency-vehicle livery or detects the flashing beacon as a temporal light signal. Transferable visual encoders such as CLIP [11] are useful for candidate recognition, but the distinction between what a vehicle *is* and what it *is doing* is central to the design adopted here (Section VI).

The literature on operator interfaces is smaller but consistent on one point that shaped this design: the dominant failure mode is not visible failure but continuing to display a stale value with no indication that it is no longer live. Endsley’s model of situation awareness [12] names this directly.

III. SYSTEM ARCHITECTURE

The system is a single measurement chain with defined interfaces cut through it (Fig. 1). A frame arrives from the camera; the detector and tracker turn it into confirmed vehicles; the speed estimator converts their motion into kilometres per hour on the road plane; the congestion engine condenses those measurements into one bounded index and a stable level; and that measurement leaves the node three times over—as annotated video and a JSON snapshot for the operator, as an archive row, and as the input the actuation subsystems act on.

Three architectural decisions make the integration work. *One owner per resource*: the node owns the camera, the model and the video source, which removes contention and allows the interface and the camera to reside on different machines. *One definition per quantity*: each measured value is computed in exactly one place and published, so two views of one junction cannot disagree at the same instant; the annotated video is rendered by the node and sent as finished frames, and the archive records the numbers that were published live. *Narrow, agreed interfaces*: the endpoint set and telemetry schema were fixed early and kept small, allowing each subsystem to be developed and tested against the others without either side breaking.

Video is transported as MJPEG over HTTP, whose per-frame independence means a dropped frame costs one frame rather than a group of pictures. Telemetry is JSON over HTTP

TABLE I
COMPARISON OF REPRESENTATIVE VISION-BASED TRAFFIC MONITORING AND SIGNAL CONTROL WORKS

#	Work	Year	Method	Accuracy	Control loop	Principal limitation
1	Pudaruth <i>et al.</i> [13]	2024	YOLOv8	96.1%	No	Detection only; no signal-optimisation layer
2	Patil <i>et al.</i> [14]	2023	Deep learning + CV	Not stated	Yes	Simulation only; no real-world validation
3	Ravichandran <i>et al.</i> [15]	2024	YOLOv8	High	Yes	Single intersection; no multi-junction coordination
4	Bindu Madhavi <i>et al.</i> [16]	2023	Deep learning	Not stated	No	Bandwidth heavy; no edge processing
5	Govinda Rao <i>et al.</i> [17]	2022	Machine learning	Not stated	No	No adaptive signal response
6	Ubaid <i>et al.</i> [18]	2022	Deep learning + CV	High	Yes	Single intersection type; ignores pedestrian flow
7	Liu <i>et al.</i> [19]	2021	Edge computing + CV	High	No	Hardware cost high; no live analytics interface

None of the seven reports vehicle speed in engineering units, and none publishes its measurement to an operator interface in real time.

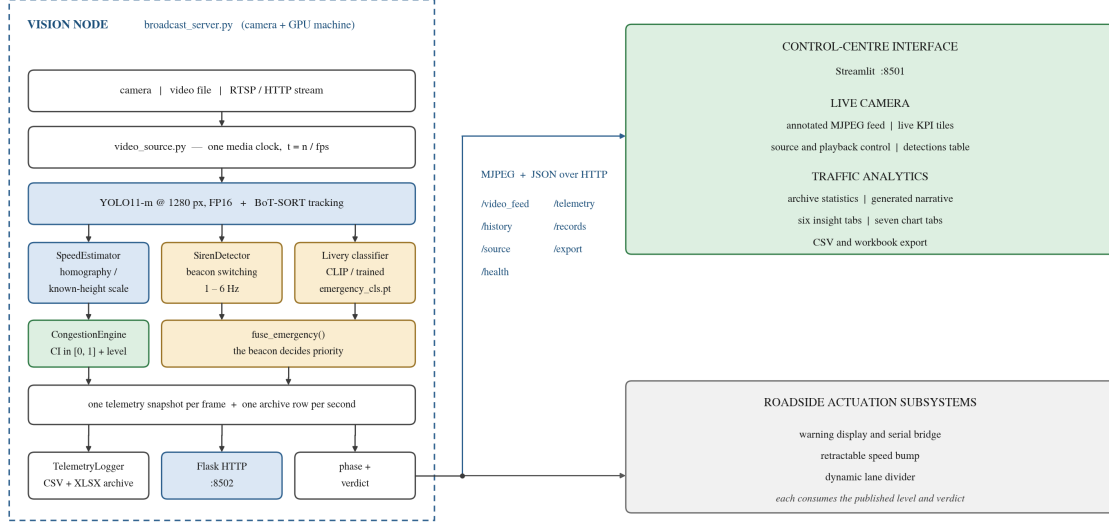


Fig. 1. System architecture. The vision node owns the camera, the detector and the video source, and publishes one telemetry snapshot per frame and one archive row per second. The control-centre interface and the roadside actuation subsystems are both clients of that published measurement; neither re-derives any quantity the node has already computed.

polled at 1 Hz, with a 4 s request timeout and a 6 s staleness threshold. The immediate log is CSV appended once per second, since an append-only file cannot be corrupted by a crash mid-write; the presentation log is an XLSX workbook written atomically. The controller chain is reached by a pipe-delimited serial protocol at 9,600 Bd.

IV. VISION SUBSYSTEM

A. Detector and Resolution Selection

Every selection reported here was made by measurement on the deployment footage rather than by published benchmark, because published accuracy figures describe the dataset a model was evaluated on and not the road this system watches. Eleven detector and input-resolution combinations were benchmarked over 60 consecutive frames of 1080p night footage on an NVIDIA RTX 4050 Laptop GPU (6 GB) under FP16 precision (Table II); the vehicles-per-frame column is the mean number of *tracked* vehicles per frame.

Three findings follow, each contradicting an initial assumption. *Resolution buys detections; model capacity does not*: every family gains about five vehicles per frame from 640 to

TABLE II
DETECTOR BENCHMARK: 60 NIGHT FRAMES, FP16, RTX 4050 LAPTOP GPU

Model @ input size	ms/frame	p95 (ms)	FPS	Veh./frame
YOLO11-s @ 640	14.2	16.0	70.6	18.4
YOLO11-s @ 960	16.0	18.7	62.5	20.8
YOLO11-s @ 1280	18.4	20.4	54.3	22.8
YOLO11-m @ 640	17.2	19.3	58.0	17.0
YOLO11-m @ 960	19.0	21.8	52.8	19.3
YOLO11-m @ 1280	27.3	29.4	36.6	23.0
YOLO11-l @ 640	22.5	24.9	44.4	17.9
YOLO11-l @ 960	24.1	26.4	41.6	19.8
YOLO11-l @ 1280	32.8	35.5	30.5	22.5
YOLO12-s @ 1280	24.0	25.6	41.7	23.6
YOLO12-m @ 1280	38.7	41.9	25.8	22.1

The selected operating point is shown in bold.

1280 pixels, while at fixed resolution YOLO11-l finds *fewer* vehicles than YOLO11-m (22.5 against 23.0) for 20% more computation. The binding constraint is how many pixels a distant vehicle occupies. *A newer generation was measurably worse*: YOLO12-m @ 1280 is the slowest configuration

tested and finds fewer vehicles than YOLO11-m, so a 33% throughput improvement and a higher yield were obtained by moving *backwards* one generation. *Half precision pays for the resolution*: FP16 approximately doubles throughput on the tensor cores of the development GPU, without which 1280-pixel input at this model size would not sustain the required frame rate.

One question is recorded rather than resolved. YOLO11-s @ 1280 reaches within 1% of the selected yield at 48% greater throughput, but this is deliberately not acted upon: vehicles-per-frame is a count, not a correctness score, and with no hand-labelled boxes a model drawing *more* boxes cannot be distinguished from one drawing *better* ones.

BoT-SORT [20] was selected over ByteTrack [21] and the earlier SORT family because identity-switch robustness governs speed validity: a swapped identity produces a large spurious displacement and therefore a large spurious speed. Its global motion compensation gives additional tolerance to a camera nudged during operation. The underlying association model in all of them is the Kalman filter [22].

B. Perspective Projection and the Metric Problem

Under the pinhole camera model, a world point (X, Y, Z) projects to image coordinates

$$x = f \frac{X}{Z}, \quad y = f \frac{Y}{Z} \quad (1)$$

where f is the focal length in pixels. Differentiating with respect to time for motion parallel to the image plane gives

$$\frac{dx}{dt} = \frac{f}{Z} \cdot \frac{dX}{dt} \quad (2)$$

Equation (2) is the formal statement of why pixel velocity is not physical velocity: image velocity is scaled by f/Z , and depth Z varies continuously across a road scene. Fig. 2(a) shows the consequence—the same 10 m of road subtends 190 pixels near the camera and 62 pixels at range.

C. The Planar Constraint

The constraint used here is that road vehicles are supported by a plane. For points on a plane, the projective relationship between world and image coordinates collapses to a homography [23]:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \mathbf{H} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}, \quad x = \frac{u}{w}, \quad y = \frac{v}{w} \quad (3)$$

where $\mathbf{H}$ is a 3×3 matrix defined up to scale, with eight degrees of freedom, recoverable from four point correspondences. Because (3) carries no depth term, it removes exactly the unknown that makes (1) unusable for metric work. Calibration consists of the operator selecting four road points forming a rectangle of known real dimensions—in the deployed calibration, a patch 14.0 m by 97.0 m. Thereafter any image point on the road plane maps to metric coordinates.

This yields the most consequential geometric decision in the pipeline. The homography is valid only for points on the plane, and a bounding-box centroid floats in mid-air at bumper

height. The bottom-centre of the box—the tyre contact patch—is the one point on a vehicle that lies on the road surface, and it is the point the pipeline tracks (Fig. 2(b)). Using the centroid would bias tall vehicles systematically relative to cars, and the bias would vary with range.

D. Uncalibrated Fallback

Where no calibration exists, scale is recovered from known object dimensions. If a vehicle of class c has typical real height H_c and bounding-box height h pixels, the local scale in metres per pixel is

$$s = \frac{H_c}{h} \quad (4)$$

Height is used rather than width because it is nearly invariant to heading: a car viewed from the side subtends roughly 4.5 m horizontally but still about 1.5 m vertically. Each vehicle supplies its own scale at its own image position, so perspective is compensated implicitly. Assumed heights are 1.50 m for a car, 1.60 m for a motorcycle and rider, 3.20 m for a bus and 3.50 m for a truck. The mode has a documented failure case: for a near-vertical camera the box's vertical extent is filled by the vehicle's ~ 4.5 m *length* rather than its height, so (4) underestimates scale roughly threefold and every reported speed is too low. A two-point fixed-scale calibration is provided for such views, rather than silently producing a plausible wrong answer.

E. The Measurement Clock and Window

Speed is displacement over time, and the choice of clock is not arbitrary. For a live camera the wall clock is correct; for a recorded video it is not. A 30 fps clip processed at 20 fps stretches each interval by 50% and reports speeds a third too low, and processing faster than real time inflates them—reported speed becomes a property of the computer rather than of the traffic. A file-backed source is therefore given its own *media clock*, advancing by exactly one frame period regardless of processing rate:

$$t_n = \frac{n}{f_{\text{source}}} \quad (5)$$

where f_{source} is the source frame rate. The verification consequence is reported in Section IX-C: the same clip played at four rates must return the same speed, and does.

Displacement is measured over a fixed 0.8 s window rather than between consecutive frames, because bounding-box regression jitters by a few pixels frame to frame and differentiating over one frame amplifies that jitter into apparent motion. For zero-mean jitter of standard deviation σ over N frames of period Δt , the standard error of the velocity estimate is

$$\sigma_v = \frac{\sigma \sqrt{2}}{N \Delta t} \quad (6)$$

so noise attenuates approximately as $1/N$ while the window stays short enough to resolve genuine acceleration; at the operating frame rate 0.8 s contains 18–24 samples. A minimum of 0.25 s and three samples is enforced, so a vehicle that has just appeared reports no speed rather than a noisy one, and readings above 200 km h^{-1} are discarded as tracker identity switches. The result is smoothed by an exponential moving average with $\alpha = 0.35$.

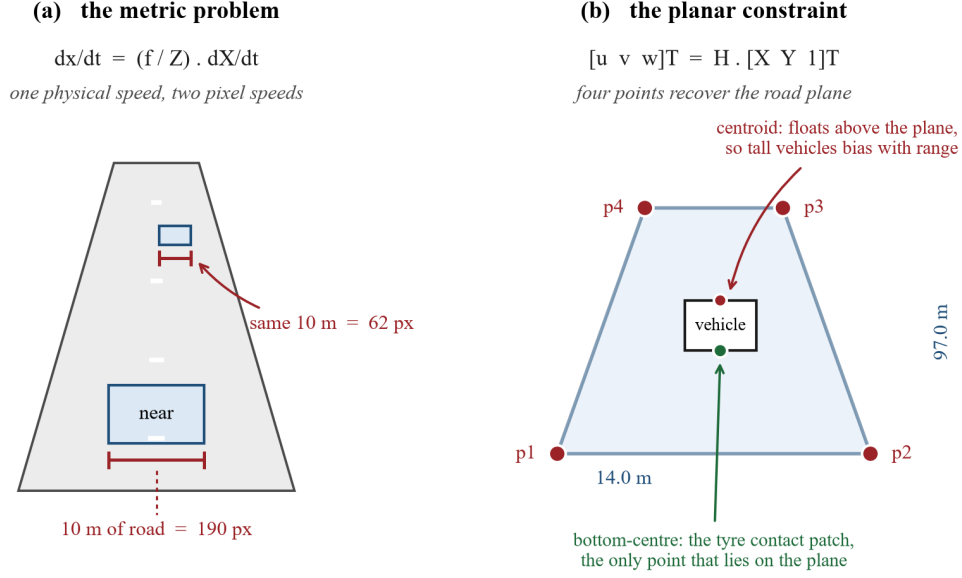


Fig. 2. The metric problem and the planar constraint that resolves it. (a) The same 10 m of road subtends a very different number of pixels near the camera and far from it, so pixel displacement per second is not a speed. (b) Four points of known real dimensions recover a homography; the bounding box's bottom-centre point is the only point on the vehicle that lies on the plane for which that homography is valid.

V. CONGESTION INDEX AND LEVEL MAPPING

A single bounded scalar is required, because nothing downstream can act on a vector of per-vehicle measurements: a control decision, a chart axis and a status tile all need one number. Following the speed–density basis of [5] and [6], the index combines a speed term and a density term:

$$CI = \text{clamp}_{[0,1]} \left(w_s \left(1 - \frac{v}{v_f} \right) p + w_d \frac{D}{D_{\max}} \right) \quad (7)$$

where v is mean measured speed, v_f the free-flow reference (50 km h^{-1}), D the PCE-weighted density, $D_{\max}$ the weighted count treated as a full road (20 PCE), and $w_s = 0.60$, $w_d = 0.40$. Both terms are individually clamped, so $CI \in [0, 1]$. PCE weights are 1.0 for a car, 0.5 for a motorcycle and 3.0 each for a bus and a truck.

The term p is a *presence factor*,

$$p = \min \left(1, \frac{N}{N_{\min}} \right), \quad N_{\min} = 3 \quad (8)$$

where N is the number of tracked vehicles. Equation (8) prevents a single slow vehicle on an otherwise empty road from registering as congestion: one slow driver is one slow driver, whereas congestion requires enough vehicles to be collectively slow. A second guard sets $CI = 0$ whenever $N = 0$ —a first implementation retained the last observed speed and could report heavy congestion with nothing in frame. A third case is easy to overlook: vehicles can be counted before any has been tracked long enough for a speed to be measured, and returning the density term of (7) alone for that interval is what stops the index reporting free flow on a road that is visibly filling up. Algorithm 1 states the implemented function with all three guards.

Algorithm 1 Congestion index with presence and emptiness guards

```

1: Input:  $\bar{v}$  mean speed (may be undefined),  $D$  PCE-weighted density,  $N$  vehicle count
2: Output:  $CI \in [0, 1]$ 
3: if  $N \leq 0$  then
4:   return 0.0 {empty road is free-flowing}
5: end if
6:  $d \leftarrow \text{clamp}_{[0,1]}(D/D_{\max})$ 
7: if  $\bar{v}$  is undefined then
8:   return  $d$  {present, not yet measurable}
9: end if
10:  $p \leftarrow \text{clamp}_{[0,1]}(N/N_{\min})$ 
11:  $s \leftarrow \text{clamp}_{[0,1]}(1 - \bar{v}/v_f)$ 
12: return  $\text{clamp}_{[0,1]}(w_s \cdot s \cdot p + w_d \cdot d)$ 

```

The continuous index is mapped onto three levels—FREE, MODERATE and HEAVY—at thresholds 0.30 and 0.60. A bare comparator oscillates whenever its input sits on a threshold, so the mapping applies hysteresis of ± 0.05 with a minimum dwell of 1.5 s, functionally a Schmitt trigger. Since the level is published and acted upon by physical actuators, an index hovering near 0.60 would otherwise flip the reported state at frame rate.

VI. EMERGENCY VEHICLE RECOGNITION

A. Livery Recognition and a Statistical Failure

Two livery recognisers were evaluated: a classifier trained on project footage, and zero-shot CLIP ViT-B/16 [11]. Table III reports both on cameras neither had seen and on the camera the classifier was trained for.

TABLE III
LIVERY RECOGNISER COMPARISON

Criterion	CLIP	Trained classifier
<i>On cameras neither recogniser had seen</i>		
Daylight confidence	ambulance, 0.95 (found)	0.55 (missed)
Ordinary night cars flagged	0 of 8	6 of 8, at 0.94–1.00
False alarms in 4186 daylight detections	0	—
<i>On the camera the classifier was trained for</i>		
Ambulance recognised (> 0.70)	53.4%	71.1%
Police car recognised	41.1%	71.4%
Fire engine recognised	39.7%	70.3%
False alarms, this camera	0.0%	0.0%
False alarms, unrelated imagery	0.3%	5.3%
Cost per crop	4.3 ms	0.7 ms

The trained classifier is more accurate and about seven times cheaper on the camera it was built for, and is retained as a selectable option for that case. CLIP is the general-purpose default because it is the option that travels: a deployed system is pointed at cameras for which nobody has training data, and a recogniser flagging six of eight ordinary cars is not merely less accurate but harmful, since every false alarm proposes an unwarranted intervention in live traffic.

A systematic fault in the first livery implementation is reported here because it was found by measurement rather than inspection, and because it motivated the design change of Section VI-B. That implementation summed all 14 emergency prompt scores against the 15 ordinary ones. Softmax spreads probability across all prompts, so an uninformative crop produces a near-uniform distribution in which 14 of 29 prompts sum to approximately 0.48—above the 0.45 decision threshold. Every unreadable crop was therefore classified as an emergency vehicle: 88 of 299 vehicles on the night footage, or 29% of all traffic. The flagged crops shared darkness, not appearance. Restructuring the comparison as best-emergency against best-ordinary drops the same crop to 0.25, and a readability gate, three agreeing observations and asymmetric thresholds were added as safeguards. The flaw was statistical rather than a coding error, which is why only measurement exposed it.

B. Beacon Detection as a Temporal Signal

A beacon is defined by its behaviour in time, not by its appearance in one frame. A red pixel region is a tail lamp; a red pixel region that *switches* is a beacon. The detector evaluates four conditions over a 3.4 s observation window.

Spatial gate. Only the upper portion of the bounding box is examined, extended 18% above the roofline because a bright lamp blooms beyond the metalwork it sits on. This also excludes brake lights, which are red and located low on the body.

Chromatic purity. For red, $G < 0.55R$; for blue, $R < 0.70B$; with luminance at least 165/255. The threshold is empirically grounded: on the night footage genuine beacon pixels occupy 200–255 while bluish paintwork and reflections occupy 130–155, so 165 separates the populations.

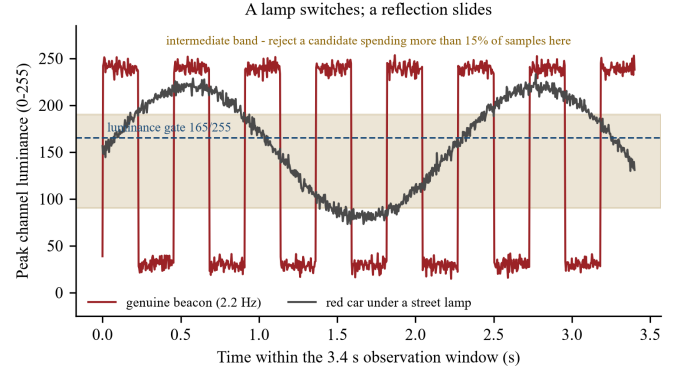


Fig. 3. The discriminating test. A genuine beacon is a two-state device and is almost never sampled between states; a reflection sliding across bodywork spends most of its time in the intermediate band. Measured on the night footage, the real beacon scored 0.00 on the intermediate fraction while every false positive scored between 0.18 and 0.43.

Rate plausibility. At least three cycles, at 1–6 Hz. Beacon flash rates are legally regulated; below 1 Hz is not a beacon and above 6 Hz is sensor noise.

Switching character. This is the discriminating test. A lamp is a two-state device—fully on or fully off—and is almost never sampled mid-transition. A red car passing beneath a street lamp brightens and dims *continuously* and spends much of its time at intermediate luminance. The detector computes the fraction of samples in the intermediate band and rejects a candidate above 15% (Fig. 3). It separates the two cases by a *physical property of the light source* rather than by a tuned threshold on appearance.

C. Fusion Hierarchy

The two recognisers are fused with a deliberate hierarchy. A confirmed flashing beacon yields priority and justifies intervention; livery recognition alone yields a named but non-priority flag, displayed to the operator and not acted upon. Where the beacon is blue the vehicle is police; where it is red, the livery classifier disambiguates ambulance from fire appliance, since only the markings separate those two.

VII. OPERATOR INTERFACE AND INTEGRATION

The control-centre interface is a multi-page browser application in Python, selected so that the interface and the vision pipeline share one language, one dependency set and one test runner. It presents live and historical views over a network boundary and may run on a machine separate from the camera. Its layered structure separates presentation from a service layer, typed domain models and interchangeable simulated and live providers, so the data source can change without changing every widget; persistent state uses a local SQLite schema.

Fig. 4 shows the Live Camera page. Every quantity the operator acts on is visible on one screen and is the quantity the node published: per-vehicle class and speed in km/h drawn on the frame by the node itself, the congestion index and level, the frame rate, and—critically—the active calibration mode printed beside the speeds it produced.

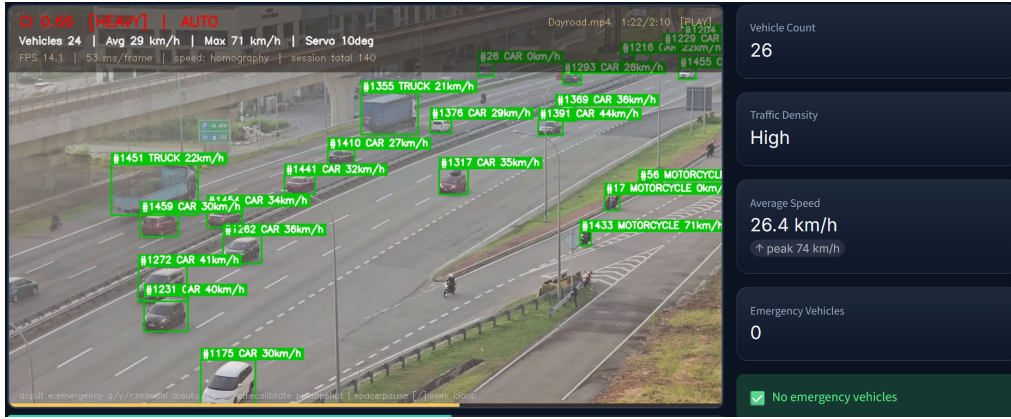


Fig. 4. The Live Camera page during a recorded-source run. The node draws per-vehicle class and speed onto the frame it publishes, so overlay and telemetry cannot disagree; the header states the congestion index and level, the frame rate, and the active calibration mode (speed: homography). The right-hand panel is the same telemetry snapshot rendered as operator-facing tiles.

What otherwise distinguishes the interface concerns trust rather than appearance, and three behaviours were added in response to observed failures rather than to a specification. *Staleness detection*: when a video source reached its end, the node kept answering telemetry requests with its final snapshot, so the interface displayed a count, a speed and an index—all frozen, none current, with nothing to indicate a problem; telemetry older than 6 s is now marked stale, because reachability is not liveness. *Labelled fallback*: when the node is unreachable the interface falls back to a clearly labelled simulated feed and marks the node offline. *Read-back after command*: after a playback or source control is used, the interface reads the node’s state back rather than assuming the command took effect.

VIII. ROADSIDE ACTUATION SUBSYSTEMS

Three actuation subsystems consume the published congestion level and emergency verdict on a scale demonstrator (Fig. 5). They are summarised here at the level needed to interpret the results of Section IX.

A. Warning Display and Serial Bridge

An ESP32 drives a 60-pixel WS2815 addressable strip, a 16×2 I²C character display and a tone-capable buzzer, and bridges the control-centre link to the downstream controller chain. It was selected over an 8-bit alternative on interface count rather than processor speed: the firmware simultaneously uses USB serial at 115,200 Bd, a second hardware UART at 9,600 Bd, I²C, an addressable-LED output and a buzzer output, which a single-UART board cannot provide without sharing a port.

Because the board is a 3.3 V device driving 5 V and 12 V peripherals, the interface circuit includes a 74AHCT125-class logic buffer for the strip data line, a 330 Ω to 470 Ω series resistor near the source, a bidirectional open-drain level shifter on the I²C bus, a resistive divider on the UART receive line, approximately 1,000 μ F of bulk capacitance across the strip supply, and a common ground across all domains. Supply sizing follows $I_{\text{design}} = NI_{\text{pixel,max}}k$ with margin k , taken from

the measured full-white current of the installed reel rather than a generic figure. The firmware maintains local warning outputs while a downstream operation is pending, so a mechanical timeout does not leave an emergency warning frozen or silent; a device-present flag suppresses display writes if the character display is absent while LED and buzzer operation continues, replacing an earlier approach in which a display initialisation failure halted the subsystem.

B. Retractable Speed Bump

An Arduino Uno drives a servo-actuated retractable bump, senses approaching vehicles with an HC-SR04 ultrasonic module, and receives emergency notifications over a 433 MHz LoRa link from a separate transmitter representing the emergency vehicle. Distance follows from acoustic time of flight,

$$d = \frac{t_{\text{echo}} c_{\text{air}}}{2} \quad (9)$$

where t_{echo} is the round-trip echo time. Equation (9) is evaluated with $c_{\text{air}} \approx 0.034,3 \text{ cm } \mu\text{s}^{-1}$ against a 20 cm decision threshold, a prototype control parameter rather than a full-scale road design distance.

The controller resolves competing requests through a deterministic three-level priority—emergency, then congestion, then automatic—so a lower-priority sensor event cannot re-raise the bump during an emergency interval. The SX1278 radio [24] is configured by direct register access at 433 MHz carrier, 125 kHz bandwidth, spreading factor 7, coding rate 4/5, eight-symbol preamble, sync word 0x12 and CRC enabled. A valid emergency packet lowers the bump for a 10 s override interval while distance measurement continues.

C. Dynamic Lane Divider

A movable divider representing a reversible-lane boundary [25] is driven by two servos through 3D-printed crank-and-link mechanisms in PLA [26]. The conversion from rotary to lateral motion follows the slider-crank relationship

$$x(\theta) = r \cos \theta + \sqrt{l^2 - r^2 \sin^2 \theta} \quad (10)$$



Fig. 5. Scale-hardware evidence. (a) The warning display driven by the ESP32, showing the addressable strip and the character display during an ambulance event. (b) The retractable speed bump installed in the scale carriageway. (c) Serial evidence of the priority hierarchy: the controller reports automatic mode, receives a valid one-byte LoRa E packet at -132 dB m with 6.5 dB SNR, activates emergency mode and lowers the bump, then continues reporting distance while the override holds. (d) The two-servo lane-divider linkage at a calibrated endpoint. These establish scale-prototype function, not roadworthiness or service life.

where r is the effective crank radius, l the connecting-link length and θ the crank angle. The prototype is not an ideal slider-crank—two separated actuation points, finite joint clearances and a long divider—so (10) explains the motion principle rather than claiming the final displacement was analytically validated. The transmitted force is likewise below the first-order estimate $F \approx \tau/r$ for servo torque τ , because linkage angle, friction, clearance and the flexibility of printed parts introduce losses; the endpoints were therefore tuned on the assembled mechanism rather than from nominal geometry.

Because the two mechanisms are mounted in opposite orientations, one servo angle increases as the other decreases. The calibrated endpoints are $50^\circ/120^\circ$ for the right position and $110^\circ/70^\circ$ for the left, giving 60° of travel on one actuator and 50° on the other. The asymmetry is a consequence of the real mounting geometry, not an error; forcing identical ranges increased skew and mechanical stress. Motion is commanded in 1° steps to reduce jerk. Servos and the addressable strip are powered separately from the controller regulator, with all grounds commoned.

IX. EXPERIMENTAL RESULTS

A. Evidence Boundaries

Evaluation was divided into four classes of evidence, reported explicitly because the division prevents a passing software test from being read as physical reliability, or unlabelled road footage as detector accuracy. Table IV states what each establishes and what it does not.

Six datasets were used: daylight and night footage at 1920×1080 for detection-yield and per-stage timing; elevated highway footage at 1280×720 as the live demonstration source and the origin of most of the archive; a mixed emergency-vehicle clip for the recognition and priority path; an 11 min 43 s public compilation from an unseen camera (21083 frames) as a generalisation stress test; and synthetic clips in which a target travels at exactly 18, 36, 72 and 16.2 km h^{-1} .

TABLE IV
EVIDENCE CLASSES AND CLAIM BOUNDARIES

Evidence class	Supports	Does not support
Constructed ground truth	Correctness of the geometry, media clock and speed arithmetic	Field speed accuracy under real calibration error and occlusion
Recorded road archive	Long-run behaviour of the index; class mix; throughput	Detection accuracy; any travel-time or delay benefit
Automated regression	Interface, schema, endpoint and logic correctness	Physical reliability of any actuator
Bench hardware tests	State transitions, priority order and threshold behaviour at scale	Roadworthiness, service life, outdoor radio range

The synthetic clips deserve note. Real traffic footage has no ground-truth speed, so a measurement against it can only be compared with another estimate; by generating footage whose target velocity is constructed and therefore known exactly, the speed pipeline can be validated in the strict sense. It is the only point in the system where genuine ground truth exists.

Verification is otherwise automated: a regression suite of 833 checks across seven files covers the vision pipeline, the telemetry schema and every HTTP endpoint, video sourcing and playback, the analytics datasets, and interface rendering. Two techniques within it are method rather than result. *Synthetic clocks*: every time-dependent behaviour is asserted against an injected clock rather than by sleeping, so a test can assert that the congestion level holds for exactly its 1.5 s dwell. *Constructed motion*: speed is asserted against scripted rather than observed motion, converting a plausibility check into a correctness check.

B. Speed Accuracy Against Constructed Ground Truth

Table V reports the one place in the system where genuine ground truth exists.

Agreement is exact because the geometry and timing chain used the same controlled world coordinates as the test gen-

TABLE V
SPEED VALIDATION AGAINST CONSTRUCTED GROUND TRUTH

Calibration mode	Expected (km/h)	Measured (km/h)
Homography	18	18.00
Homography	36	36.00
Homography	72	72.00
Automatic (known height)	18	18.00
Fixed scale	18	18.00
Parked vehicle	0	0.12

erator; this is a correctness test of that chain, not a field-accuracy estimate. The parked-vehicle row is the informative one: a stationary vehicle measures 0.12 km h^{-1} rather than exactly zero, that residual being the bounding-box jitter of (6) surviving the 0.8 s window and the smoothing. It is well below the 3 km h^{-1} stopped threshold, so it never changes a reported quantity, but it is the measured noise floor of the speed pipeline and is reported rather than rounded away.

C. Media-Clock Independence

A vehicle scripted at 16.2 km h^{-1} through a real MP4 file was decoded at four rates: real time, double speed, half speed and unthrottled. All four returned 16.2 km h^{-1} , agreeing to two decimal places. This confirms that reported speed is a property of the traffic and not of the machine, and is the direct experimental consequence of (5).

Field accuracy is stated as $\pm 5\%$ calibrated and $\pm 25\%$ automatic. These are the calibration modes' own error bounds, not a measured field error, which would require instrumented reference vehicles that were not available. Field error would additionally include calibration error, the contact-point approximation, occlusion, tracking switches and non-planar road effects.

D. Throughput

Per-stage cost was measured on day and night clips at the default configuration. Detection dominates at two-thirds to four-fifths of the frame budget, which identifies it as the only stage where optimisation would return anything measurable. Both recognisers cost more at night, for different reasons: the livery classifier because dark crops fail the readability gate and are retried rather than backed off, and the beacon detector because at night there are genuinely beacons to measure. During the live demonstration the node reported 23.1 fps averaged over $3,988 \text{ s}$ of continuous running, against 22.3 fps predicted for night footage and 28.5 fps for day.

Two instrumentation findings are recorded. First, a re-measurement of the default configuration on a later date returned 31.7 ms against the 27.3 ms of Table II—about 16% slower—while the yield was identical at 23.0 vehicles per frame, establishing that only the clock moved. Table II therefore remains valid for comparison between settings, since every row would shift together. Second, a frame rate below the benchmark prediction was traced by three controlled runs on identical footage: 8.8 fps with the preview window open at 1280 px , 11.4 fps with it closed, and 11.9 fps closed at 960 px .

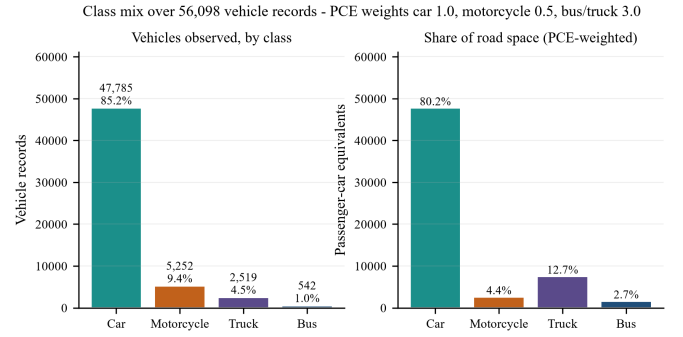


Fig. 6. Class mix across 56 098 vehicle records (left) and the same mix re-expressed as passenger-car equivalents (right). Cars are 85.2% of vehicles but 80.2% of road space; trucks are 4.5% of vehicles but 12.7% of road space.

The suspected cause—disk writing inside the frame loop—was formally excluded, since workbooks are written on a timer and at shutdown.

The preview window therefore costs about 30% of throughput, while dropping resolution to 960 buys only 0.5 fps and loses roughly 25% of the vehicles found at night; the correct action was to keep the resolution and run headless, observing through the browser interface. One caution about instrumentation follows: reported inference time itself fell from 58.7 ms to about 50 ms at *identical* resolution once the window closed, because the window's event polling inflates the measurement.

E. Field Measurements

The results below are drawn from the whole archive—26 071 logged seconds (7.24 hours) across 94 sessions recorded between 8 and 20 August 2026, containing 56 098 individual vehicle records of which 46 228 carry a measurable speed. Of the telemetry rows, 25 861 (99.2%) were recorded under homography calibration. Nothing was selected after the fact.

Over the 46 228 speed-carrying records the median mean-track speed is 30.2 km h^{-1} with an interquartile range of 22.2 km h^{-1} to 38.1 km h^{-1} . The distribution is unimodal with a mode just below the 50 km h^{-1} free-flow reference, a spike at the bottom for stopped traffic, and a thin tail rather than a spread of implausible values. Scaled pixel displacement would not produce this shape.

Fig. 6 reports the class mix across all 56 098 records and the same mix re-expressed as road space consumed. It is the clearest justification for PCE weighting the study produced: trucks are 4.5% of the vehicles observed but 12.7% of the road space they occupy, a nearly threefold difference. A metric counting vehicles rather than weighting them would under-report exactly the traffic that causes congestion.

F. Behaviour of the Congestion Index

Fig. 7 plots mean measured speed against the congestion index over every logged second in which vehicles were present and a speed was measurable (17 661 of the 26 071 rows). Mean speed falls in every successive band, from 68.3 km h^{-1} in the

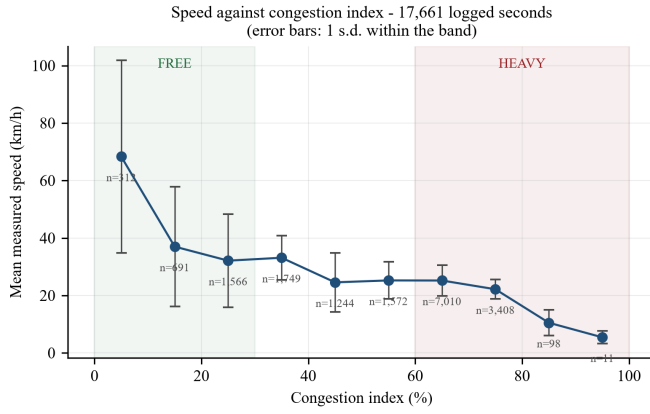


Fig. 7. Mean measured speed against the congestion index over 17 661 logged seconds in which vehicles were present and a speed was measurable. Error bars are one standard deviation within each band; sample counts are annotated. Mean speed falls monotonically from 68.3 to 5.4 km h⁻¹ across the index range.

TABLE VI
EMERGENCY RECOGNITION ON UNSEEN FOOTAGE

Measure	Value
Frames analysed	21 083
Scene cuts detected	301
Vehicles tracked	2 511
Recognised as emergency	166 (6.6%)
Distinct vehicles, identity churn merged	≈96
Beacon-confirmed, i.e. given priority	21
By type	60 fire, 59 ambulance, 47 police
Confidence range	mostly 0.93–1.00
Beacon rates measured	1.2 Hz to 4.8 Hz

lowest to 5.4 km h⁻¹ in the highest—a decline of 92% across ten bands and tens of thousands of samples.

This result should be read precisely. Speed is itself an input to (7) with weight 0.60, so a negative association between index and speed is expected by construction and Fig. 7 is not an independent validation dataset. What the figure adds is that the ordering survives everything that could break it—the density term, the presence gate, the emptiness guard and the hysteretic level mapping—across the full index range and at band populations from 11 to 7010 samples, and that the resulting curve has the shape and magnitude the classical speed–density relationship predicts [6]. That is consistency with traffic theory on the system’s own field data, which is a weaker claim than independent validation and a stronger one than internal arithmetic. Across the archive, logged time divides as 41.9% FREE, 18.4% MODERATE and 39.7% HEAVY.

G. Emergency Recognition

Table VI reports a full run over the 11 min 43 s compilation from an unseen camera, with no region of interest and no selection of favourable segments.

The 6.6% figure is a *selectivity* figure and is stated as one. It is not an accuracy figure and not a false-positive rate; establishing either would require labelling all 166 crops by hand. A sample was inspected and most were unmistakable,

TABLE VII
BENCH OBSERVATIONS FOR THE THREE ACTUATION SUBSYSTEMS

Subsystem / test	Observed result
Warning controller	Strip, character display and buzzer produced the commanded state; local output continued during downstream waits
Bump, clear path	54 cm to 56 cm measured; bump lowered
Bump, near object	15 cm to 16 cm measured; bump raised
Congestion override	Raised at both 56 cm and 18 cm to 19 cm, i.e. independent of the ultrasonic reading
Emergency packet	Valid E command lowered the bump for 10 s while distance continued to update
Unknown packet	Ignored; controller retained the applicable mode
Lane divider	Both calibrated endpoints reached repeatedly; 1° stepping reduced visible jerk and skew

though a few were ambiguous crops from dense traffic. What the number establishes is that the recogniser is discriminating rather than indiscriminate—a detector flagging everything would achieve perfect recall and demonstrate nothing. That the measured beacon rates of 1.2 Hz to 4.8 Hz fall inside the legally regulated band is useful independent corroboration.

That only 21 of 166 reached beacon-confirmed priority is expected rather than a defect: in daylight a beacon bright enough to register saturates the sensor and blooms to white, which correctly fails the colour-purity gates. Measured on a daylight ambulance, beacon channel score 0.04 against livery score 0.95. The two recognisers cover different conditions by design, and Fig. 8 shows the distinction in operation.

On the night footage the switching test of Section VI-B separated the populations completely: the genuine beacon scored 0.00 on the intermediate-luminance fraction, while every false positive scored between 0.18 and 0.43. Combined with making the beacon rather than the livery the priority trigger, false alarms on that footage fell from 88 of 299 vehicles (29%) to zero. The same asymmetry appears across the archive: of 277 records carrying any emergency verdict, 246 were flagged by livery with no beacon and given no priority, while 31 carried a confirmed beacon—the conservative path taken far more often than the interventionist one, as intended. This supports the decision boundary; it does not guarantee a zero false-alarm rate on other cameras, countries, weather or beacon patterns.

H. Bench Hardware Results

Table VII reports the actuation subsystems, each tested by changing one operating condition at a time and recording the controller state; the supporting captures are shown in Fig. 5.

The automatic rows verify the 20 cm threshold: the state change follows the measured distance rather than a mode change. The congestion rows verify that the override is independent of the ultrasonic reading. The emergency row is the strongest priority evidence, since the receiver recorded the packet, activated emergency mode and lowered the bump for the full override interval while distance values continued to update—and ignored an unknown packet rather than acting on it. Divider acceptance rested on observed motion, free travel, endpoint positioning and correct display and lighting behaviour; no encoder was installed, so skew is reported as visually acceptable rather than numerically measured, and no



Fig. 8. The fusion hierarchy in operation. Left: livery without a beacon—the ambulance is recognised and named, boxed in amber with a question mark, and no priority verdict is published. Right: the same pipeline seconds later with a confirmed beacon—the vehicle is boxed in magenta, the measured flash rate of 3.3 Hz is printed on the box, and a priority verdict is published. Note that the speeds in the left frame are erroneous (138 km h⁻¹ for a suburban car) because the active homography belongs to a different camera; this is the calibration hazard of Section X visible in a screen capture.

TABLE VIII
BEHAVIOUR WHEN A COMPONENT IS REMOVED

Component removed	Observed behaviour
Actuation subsystems	Node continues measuring, publishing and logging; interface unaffected
Vision node	Interface falls back to a labelled simulated feed and marks the node offline; no page errors
Video source ends	Telemetry marked stale after 6 s; with looping enabled the source restarts
Network path lost	Interface reports the node offline within one poll and switches to labelled fallback; node keeps measuring
Calibration file absent	Pipeline falls back to known-height scaling, states the mode on every frame, reports the wider error bound

cycle-life dataset exists. The linkage batch consumed 65.06 g of PLA by slicer estimate. Claims such as activation latency, sound pressure and viewing distance are not made, because the controlled measurements that would support them were not completed.

I. Integration and Degradation

End to end, the integrated system ran at 27.1 fps to 28.5 fps on day footage and 22.3 fps at night, sustaining 23.1 fps over its longest logged run. Table VIII reports its behaviour with each component removed in turn—a distributed system’s behaviour when a component is missing being as much a design property as its behaviour when everything works.

Of the 833 automated checks, 831 pass in a single run, and 833 is not reachable in one run. The first six suites pass 724 of 724 unconditionally. The interactive suite requires the vision node in two states at once: with the node down it scores 107 of 109, with the node up 105 of 109. The honest single-run result is therefore 831 of 833; resolving it means splitting that suite along the live/offline boundary so each job has an unambiguous denominator—a structural limitation of the test design, not an intermittent failure.

X. DISCUSSION

A. What the Results Establish

The strongest result is not a claimed improvement in traffic flow; it is traceable measurement. Speed, the quantity everything else derives from, returns 18.00, 36.00 and 72.00 km h⁻¹ against constructed targets and an identical 16.2 km h⁻¹ across four playback rates—a correctness check against known truth, available only because footage was constructed whose ground truth was controlled. Over 17661 measurable seconds the index orders the observed road as traffic theory predicts, across 94 sessions rather than a window chosen afterwards.

The emergency experiment demonstrates why a recognition label and an actuation authorisation should be different variables. Making a flashing beacon rather than a vehicle’s markings the trigger took false alarms on night footage from 88 of 299 vehicles to zero, and the same asymmetry persists in the archive. A camera may describe a vehicle with moderate confidence while the controller remains in a safe non-priority state until temporal evidence arrives. Because a false declaration moves real vehicles on a real road, this is a safety result and not merely an accuracy one.

The project also demonstrates modular integration. HTTP and serial contracts decouple the vision process, the browser interface and the microcontrollers, and explicit priority rules make actuator state explainable—preferable to an opaque chain in which an uncertain detector directly moves hardware. Some physical integration was nonetheless demonstrated by command rather than as a continuously autonomous field trial.

B. Sources of Error

The errors encountered fall into four classes, each requiring a different response. *Geometric* errors—a calibration applied to the wrong camera—produce confidently wrong readings, handled by refusing to carry a calibration across a source change and printing the active mode on every frame; Fig. 8 shows the failure mode when that discipline is not observed. *Statistical* errors—the 29% false-positive fault—are invisible to code review and were found by asking what the false positives had in common; the fix was structural. *Systems* errors—resource contention, a stale snapshot, a silent

misconfiguration—were resolved by making the system report its own state rather than making it less likely to fail. *Measurement* errors—16% throughput variation on an unchanged configuration—were handled by identifying an invariant that does not move with machine load, namely detection yield.

C. What the Results Do Not Establish

Table IV draws the boundaries; four consequences are worth stating outright.

No mean average precision figure is quoted, because no hand-labelled boxes exist for this footage; the detector benchmark ranks configurations by yield and cannot rank them by accuracy, which is why the 48%-faster small-model configuration was not adopted. The emergency flag rate of 6.6% is a selectivity figure, not an accuracy figure. The automatic mode's $\pm 25\%$ bound is the calibration mode's own error bound, not a measured field error. And the claim of reduced waiting time under adaptive control is *not* made: the mechanism is demonstrated—the signal serves measured demand rather than a fixed plan, and that measurement is validated—but the reduction itself was not measured, and asserting it would require a controlled before-and-after study at a live junction with instrumented vehicles.

The actuation subsystems are validated as scale prototypes: the bump and divider carry no vehicle loading, have no independent position feedback and no cycle-life data, and the radio link was exercised at bench separation only. The hourly-demand analytics can also mislead when sessions are unevenly distributed across the day, since a headline naming the busiest clock hour reflects when the system was run as much as when the road was busy.

D. Limitations and Future Work

Table IX states the principal limitations with the work that would address each; two further items—beacon saturation in daylight, and an interactive test suite that cannot pass fully in one run—are noted in Sections IX and VI. The highest-value outstanding item is labelled ground truth for detection: several hundred hand-labelled frames would convert the detector benchmark from a comparison of yields into a comparison of accuracies and settle the open small-model question.

E. Wider Considerations

Three considerations shaped design decisions rather than merely following from them. *Safety*: the vision subsystem actuates nothing directly, so its safety contribution is the quality of its measurement—priority is published only on a confirmed beacon, calibration mode is printed beside its speeds, stale data is marked and simulated data labelled. The moving linkages create pinch points, and a road-ready design would require guards, an emergency stop, obstruction detection and formal risk assessment under ISO 12100 [27], with signal-controller functional safety under EN 12675 [28]. *Whole-life cost*: an inductive loop is consumed by the road it is cut into and destroyed by the next resurfacing, whereas

TABLE IX
LIMITATIONS AND CORRESPONDING FUTURE WORK

Limitation	Future work
No labelled ground truth for detection	Hand-label several hundred frames and run a genuine mAP evaluation
Environmental benefit argued, not quantified	Controlled before-and-after study at a live junction with instrumented vehicles
Recogniser and calibration can be bound to the wrong camera	Store a source signature in the calibration file and refuse mismatches
One camera monitors one approach	Extend to several nodes feeding one interface; the schema already supports it
Actuation prototypes are open-loop; radio verified at bench distance	Add end-position sensing, obstruction detection and an emergency stop; test radio range outdoors
Command path has no authentication	Authenticated roles, append-only command records, positive actuator acknowledgement

a camera mounts on existing street furniture and survives it; the node is commodity hardware, and the archive is written in open formats so the data outlives the software. *Privacy*: the system records counts, classes and speeds, performs no number-plate recognition and retains no frames beyond the rolling stream, since a system that also identified drivers would carry a far heavier justification burden under data-protection law [29]. Beacon conventions are internationally regulated where liveries are not, which is also what allows deployment outside the fleet the system was developed against.

XI. CONCLUSION

This paper has reported a complete, measured chain from a single roadside camera to physical roadside actuators. Vehicles are detected and tracked with a detector–resolution operating point selected by benchmarking eleven alternatives on the deployment footage, a process that overturned the assumption that a newer model generation would perform better. Image motion is converted to kilometres per hour through a ground-plane homography anchored at the tyre contact patch and timed against a media clock, so that reported speed is a property of the traffic rather than of the computer; against constructed ground truth the pipeline returns 18.00, 36.00 and 72.00 km h⁻¹ for targets scripted at 18, 36 and 72, and an identical 16.2 km h⁻¹ across four playback rates.

Those measurements are condensed into a bounded congestion index guarded so that neither an empty road nor a single slow driver registers as congestion, and mapped to a hysteretic three-level state suitable for driving physical actuators; over 17 661 measurable seconds the index orders the road as traffic theory predicts. Emergency vehicles are recognised by two complementary cues, with only a confirmed flashing beacon authorising intervention—a change that reduced night-footage false alarms from 29% of all traffic to zero. The integrated system sustains 22–28 fps on a laptop-class GPU, degrades in a defined and labelled way when any one component is removed, and is covered by 833 automated checks.

The contribution is integrative and evidential. The individual techniques are mature; what is uncommon is a prototype in which every link from photons to published numbers to physical action is validated against constructed truth where such truth can be constructed, and in which the places where it cannot be constructed are stated rather than obscured.

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Naif Mohammed is reading for the B.Eng. (Hons.) degree in electronic engineering at the Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia. His contribution covered the ESP32 warning-display controller, its protected interface circuit and the serial bridge to the controller chain.

Abdelrahman Osama is reading for the B.Eng. (Hons.) degree in mechatronic engineering at the Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia. His contribution covered the retractable speed-bump subsystem: ultrasonic sensing, the LoRa emergency link and the priority controller.

Adem Becheikh is reading for the B.Eng. (Hons.) degree in mechatronic engineering at the Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia. His contribution covered the dynamic lane divider: the 3D-printed crank-link mechanism, two-servo actuation and endpoint calibration.